Exposé on Master’s Thesis

Name:	Carsten Oertel
Matr.no.:	03723658
E-Mail:	carsten.oertel@tum.de
Supervisor:	Prof. Dr.-Ing. habil. Alois C. Knoll
Industrial supervisor:	M.Sc. Antonio Gonzales Marin (Agile Robots)
Institute:	Chair of Robotics, Artificial Intelligence and Real-time Systems
Title:	SimThreadReal: Sim-to-Real Threaded Fastening with a Dexterous Hand through Force-Guided In-Hand Manipulation
Date:	July 16, 2026

---

1 Topic

Endowing robots with human-like dexterity is a long-standing goal of robotics, and one of its most compelling proofs is the ability to grasp and manipulate the objects and tools of a world built for human hands [1]. A truly general-purpose robot should not rely on a task-specific end-effector for every job: a multi-fingered hand promises to pick up objects of varying shape, hold and reorient them, and wield the very implements and parts that people handle every day. Learning has begun to turn this promise into working skills, and dexterous in-hand manipulation in particular has advanced rapidly in recent years [2]. Some of the most demanding of these skills lie in mechanical assembly, where a firm grasp of a part must be coupled to a fine, contact-rich motion that joins it to another.

Threading a nut onto a bolt is one of the most common operations in industrial assembly, yet it remains hard to automate: the parts mate through sub-millimetre thread geometry, progress is governed by friction and contact forces that are difficult to sense, and success couples a stable grasp of the fastener to a controlled turning motion of the grasped part. A system meant to perform this task without part-specific fixturing must therefore acquire and hold the grasp itself, sense contact through that grasp, and coordinate grasping with fine, rotation-coupled manipulation.

Reinforcement learning (RL) in high-fidelity GPU simulation has recently made contact-rich assembly learnable, and nut threading has been demonstrated among such tasks [3, 4]. These learned systems, however, act with a parallel-jaw gripper and begin each episode from a fastener that is already held: the policy only ever learns to turn a pre-pinched part and never has to acquire, or actively maintain, the grasp itself. Grasp acquisition and maintenance are thus left outside the learning problem.

Multi-fingered dexterous hands, by contrast, promise far more general manipulation. Learned skills now span in-hand reorientation [5, 6], tool use [7], and bimanual tasks such as twisting lids off jars [8], and robotic fastening has been studied with guided screwdrivers [9, 10]. Yet dexterous hands remain much harder to control than grippers, precisely because of the many degrees of freedom and the rich, intermittent, hard-to-sense finger–object contact they rely on [1, 2]. The intersection of the two, a multi-fingered hand that performs a contact-rich fastening task by forming the grasp and then executing the turn, is therefore largely unexplored: to the author’s knowledge, no learned policy acquires a multi-finger grasp of a fastener and then carries out the thread start and the subsequent threading process on an arm–hand system.

This thesis aims to close part of that gap. It pursues a two-stage approach. First, a privileged teacher policy is trained in simulation to reach for a nut resting on a bolt, pinch it with a multi-fingered hand, and thread it onto the bolt. This teacher is then distilled into a vision-based student policy that replaces the privileged simulator state with RGB and, where needed, RGB-D observations, and is thus in principle transferable to real hardware, following the teacher–student recipe for dexterous grasping [11]. Relative to gripper-based assembly the task adds a reach-and-grasp phase, a side-pinch of a low-friction fastener, and the requirement to turn it down the thread into a tight success window without losing the grip. The relevance is threefold. Methodologically, the task probes credit assignment across a coupled grasp-then-manipulate sequence, contact sensing on a dexterous hand, and the limits of vision-based distillation for fine manipulation. From a simulation standpoint, it stresses how faithfully GPU physics engines represent mating thread contact, penetration-free contact simulation being an open research problem in its own right [12]. Practically, it is a step toward replacing special-purpose grippers with general dexterous hands for fastening, and, in the longer term, toward transferring such skills from simulation to reality.

1.1 Research questions

The thesis pursues five research questions:

1. Can a single PPO policy learn the coupled grasp-then-thread skill on a humanoid arm–hand system in simulation, starting without a pre-formed grasp?
2. Which contact-modelling and contact-sensing choices make the task representable and learnable in the first place, from the physics backend's representation of the mating thread geometry to filtered finger–nut contact signals in place of geometric proxies?
3. How must the reward be structured so that the policy performs genuine rotation-coupled threading instead of exploiting geometric progress signals (align-and-hold, pressing to depth without rotation)?
4. How well does the learned skill generalize across nut/bolt sizes (e.g. M10 versus M16), and how far can threading depth and multi-turn rotation be pushed?
5. Can the privileged teacher be distilled into a vision-based student policy operating on stereo RGB or RGB-D observations, and which choice of resolution, depth information, and egocentric versus third-person camera placement provides the object localization that threading requires, as the basis for sim-to-real transfer?

2 Approach

The approach builds on a vendored copy of the IsaacLab *forge* task [4], itself built on *factory* [3], and carries it from the Franka parallel-jaw gripper to a robotic arm with a five-fingered dexterous robotic hand at the end-effector. It further draws on the dexterous manipulation skills of the SimToolReal approach [7], which shows high dexterity along with the ability to learn in-hand manipulation and finger gaiting on its own. Figure 1 sketches the setup. The following subsections describe the platform and robot, the task and its control interfaces, the reward and contact design, the learning method, and the distillation to a vision-based student policy.

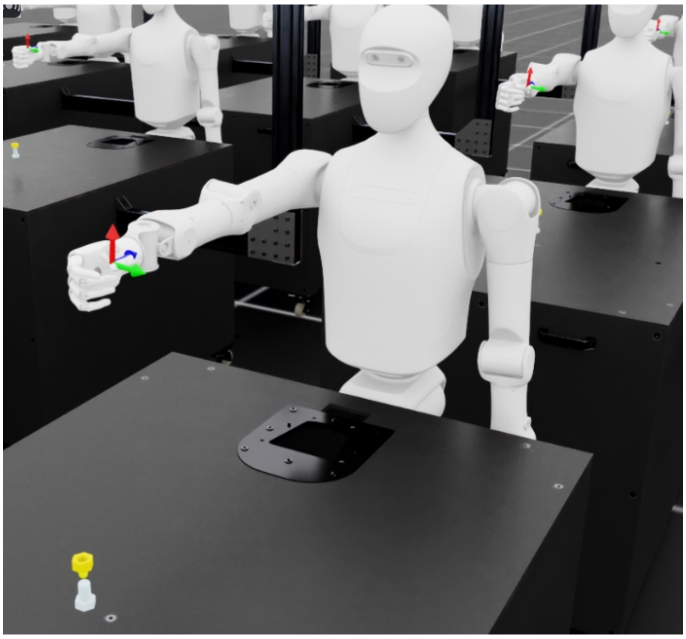

Figure 1: System overview: GPU-parallel IsaacLab environments with the arm–hand system positioned in front of the nut and bolt, driven by the staged grasp-then-thread reward.

2.1 Simulation platform and robot

Training runs in IsaacLab [13], which simulates contact-rich assembly across many environments in parallel on the GPU. Development started on IsaacLab v2.3.2 with the PhysX backend: under PhysX, the nut, modelled with a signed-distance-field (SDF) collider, could be pressed axially through the bolt thread without rotating, an exploit the policy reliably discovered, while genuinely meshing thread contact was not available as a learning signal. The collision pipeline treats the nut–bolt pair with a hydroelastic pressure-field contact model, so the mating thread geometry is physically represented: the nut and bolt thread accordingly use mesh–mesh SDF colliders, while the fingertips and the remaining robot assets are modelled as convex hulls to reduce computation.

2.2 Task and control

The nut is spawned into a manually defined pinch grip in the hand, following forge's asset-in-hand initialization, and the policy must hold this grip while threading. The 7-DoF arm is driven by a task-space impedance controller (operational-space control, $\tau = J^{\top}(\Lambda\,\dot{w}^{*} + \mu + p)$), while the fingers are driven by IsaacLab's implicit-actuator PD controller, that is, a joint-space impedance controller. Impedance control is preferred over stiff position control for two reasons: first, it improves the safety of the robot and its surroundings, since the compliant law yields on unexpected contact rather than driving through it; second, it fits a force-aware control structure, in which the policy can be conditioned on a commanded maximum contact force and the sensed contact forces, as in the force-guided forge approach [4]. Because the end-effector target is expressed relative to the bolt-tip frame, the controller draws the grasped nut toward the bolt regardless of where the nut and robot spawn, and the policy's actions probe this impedance by commanding pose-target offsets rather than direct torques. The base action space comprises end-effector pose increments (position and orientation) plus the pinch finger degrees of freedom; the yaw action is mapped unidirectionally to the tightening direction. In addition, the thesis investigates an object-centric action interface: a contact-level impedance controller through which the policy commands increments of the nut itself, namely rotation about the bolt axis, insertion depth, and grip force, and which converts these into fingertip targets. Such object-centric interfaces have proven effective for dexterous manipulation of grasped objects [7, 14].

2.3 Learning algorithm and methodology

Learning uses PPO (rl_games) in an asymmetric actor-critic setup with a privileged critic state and an LSTM, under forge-style domain randomization (control gains, thresholds, action smoothing, friction). An accompanying method study compares vanilla PPO with the mixed-exploration SAPO variant for training the teacher.

2.4 Distillation to a vision-based student policy

While the privileged teacher may exploit privileged states (the exact nut pose, contact forces), the policy that is ultimately deployed can only rely on real sensors. The thesis therefore distills the teacher into a vision-based student that replaces the privileged object state with stereo RGB and, if needed, RGB-D observations plus proprioception. The recipe follows DextrAH-RGB [11], which distils a privileged, fabric-guided grasping teacher into an end-to-end RGB student entirely in simulation through photorealistic tiled rendering. The distillation itself builds on the established interactive-imitation line, from DAgger [15] to its safety-aware variant SafeDAgger [16], which gates expert queries on a learned safety measure to make the student's on-policy data collection both more query-efficient and safer. As an alternative to fully end-to-end visuomotor distillation, the nut's position and pose could instead be supplied by a dedicated computer-vision model from NVIDIA's Isaac ROS pose-estimation pipeline, feeding the policy an estimated 6-DoF pose in place of the privileged ground truth. Either way, whether this recipe carries over to threading is open in one decisive respect: threading demands centring the nut on the bolt within about 2.5 mm, a far tighter tolerance than grasping. Which combination of higher resolution, explicit depth (RGB-D), egocentric versus third-person camera placement, and additional auxiliary supervision achieves this localization accuracy is a central experimental question of the thesis.

3 Work Plan and necessary Resources

3.1 Methodology and evaluation

The project proceeds in the five work packages of the Gantt chart (Figure 2), and the evaluation is organized along the same phases. The teacher policy (WP1–WP2) is developed through an iterative experiment cycle that changes one design decision at a time, so that the contribution of each choice (contact signal, grip shaping, spawn pose, control frame, reward weights) can be isolated by ablation. Its primary success metric is the threading success rate (nut centred, seated to target depth, and correctly turned); secondary metrics track net nut rotation, threading depth, episode length, and grip quality. The distilled student (WP3) is judged additionally by the fraction of teacher performance it retains and by the object-localization accuracy its visual observations afford across resolution, stereo versus RGB-D, and camera viewpoint. Throughout, results are compared against the pre-grasped forge baseline and across nut and bolt sizes. The final phase (WP4) attempts a sim-to-real transfer of the student policy with hardware experiments, contingent on hardware availability, before the closing documentation and writing phase (WP5).

3.2 Resources

The work requires Isaac Sim with IsaacLab in uv-managed environments. Contact-rich training runs take hours to days, and the hydroelastic SDF contact of the threading pair is computationally heavy.

The sim-to-real part additionally requires access to the physical robot hardware and corresponding supervision.

3.3 Schedule

Figure 2 shows the planned sequence in five work packages: research and baseline experiments through mid-June (WP1), one month for implementing the teacher policy via two parallel simulation pipelines and a unified pick-place-screw agent through mid-July (WP2), student distillation and transfer to our own assets through mid-August (WP3), hardware transfer and experimentation through the first week of September (WP4), and finally further experiments, documentation, and writing through mid-November (WP5). Hands-on work began in June 2026, one month ahead of the official thesis start on 1 July (dashed line).

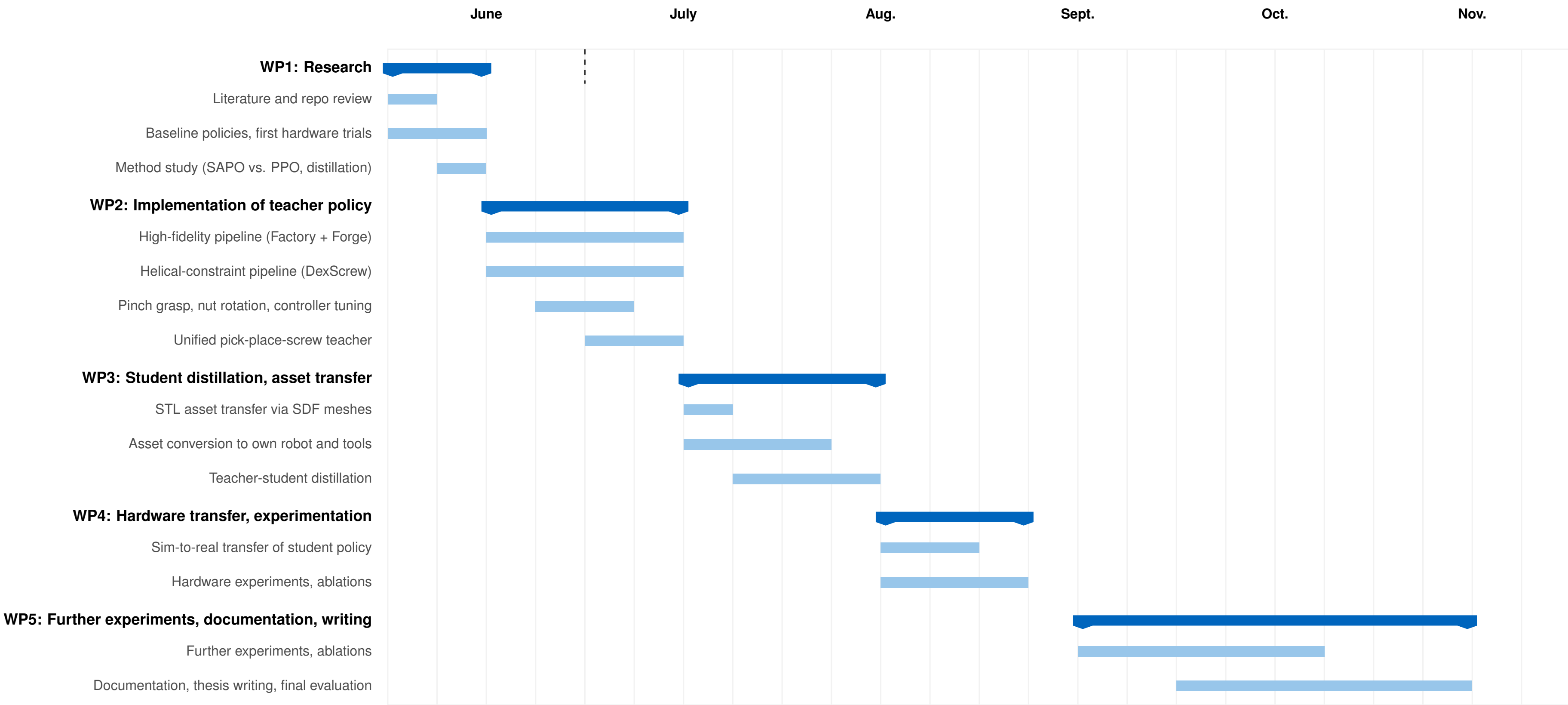

Figure 2: Time schedule (2026). The dashed line marks the official thesis start (1 July).

4 Literature

[1] Bicchi, A. and Kumar, V. “Robotic grasping and contact: a review”. In: *Proceedings 2000 ICRA. Millennium Conference. IEEE International Conference on Robotics and Automation. Symposia Proceedings (Cat. No.00CH37065)*. IEEE, 2000, pp. 348–353. ISBN: 0-7803-5886-4. DOI: 10.1109/ROBOT.2000.844081.

[2] Weinberg, A. I., Shirizly, A., Azulay, O., and Sintov, A. “Survey of learning-based approaches for robotic in-hand manipulation.” In: *Frontiers in robotics and AI* 11 (2024), p. 1455431. ISSN: 2296-9144. DOI: 10.3389/frobt.2024.1455431.

[3] Narang, Y., Storey, K., Akinola, I., Macklin, M., Reist, P., Wawrzyniak, L., Guo, Y., Moravanszky, A., State, G., Lu, M., Handa, A., and Fox, D. “Factory: Fast Contact for Robotic Assembly”. In: (May 2022).

[4] Noseworthy, M., Tang, B., Wen, B., Handa, A., Kessens, C., Roy, N., Fox, D., Ramos, F., Narang, Y., and Akinola, I. “FORGE: Force-Guided Exploration for Robust Contact-Rich Manipulation under Uncertainty”. In: (Jan. 2025).

[5] Chen, T., Xu, J., and Agrawal, P. “A System for General In-Hand Object Re-Orientation”. In: (Nov. 2021).

[6] Handa, A., Allshire, A., Makoviychuk, V., Petrenko, A., Singh, R., Liu, J., Makoviichuk, D., Wyk, K. V., Zhurkevich, A., Sundaralingam, B., Narang, Y., Lafleche, J.-F., Fox, D., and State, G. “DeXtreme: Transfer of Agile In-hand Manipulation from Simulation to Reality”. In: (Jan. 2024). URL: https://arxiv.org/pdf/2210.13702.

[7] Kedia, K., Lum, T. G. W., Bohg, J., and Liu, C. K. “SimToolReal: An Object-Centric Policy for Zero-Shot Dexterous Tool Manipulation”. In: (Feb. 2026).

[8] Lin, T., Yin, Z.-H., Qi, H., Abbeel, P., and Malik, J. “Twisting Lids Off with Two Hands”. In: (Oct. 2024).

[9] Tang, L. and Jia, Y.-B. “Robotic Fastening with a Manual Screwdriver”. In: *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, May 2023, pp. 5269–5275. ISBN: 979-8-3503-2365-8. DOI: 10.1109/ICRA48891.2023.10161139.

[10] Manyar, O. M., Narayan, S. V., Lengade, R., and Gupta, S. K. “Physics-Informed Learning to Enable Robotic Screw-Driving Under Hole Pose Uncertainties”. In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, Oct. 2023, pp. 2993–3000. ISBN: 978-1-6654-9190-7. DOI: 10.1109/IROS55552.2023.10342151.

[11] Singh, R., Allshire, A., Handa, A., Ratliff, N., and Wyk, K. V. “DextrAH-RGB: Visuomotor Policies to Grasp Anything with Dexterous Hands”. In: (Feb. 2025).

[12] Chen, A. H., Hsu, J., Liu, Z., Macklin, M., Yang, Y., and Yuksel, C. “Offset Geometric Contact”. In: *ACM Transactions on Graphics* 44 (4 Aug. 2025), pp. 1–21. ISSN: 0730-0301. DOI: 10.1145/3731205.

[13] Mittal, M., Yu, C., Yu, Q., Liu, J., Rudin, N., Hoeller, D., Yuan, J. L., Singh, R., Guo, Y., Mazhar, H., Mandlekar, A., Babich, B., State, G., Hutter, M., and Garg, A. “ORBIT: A Unified Simulation Framework for Interactive Robot Learning Environments”. In: *IEEE Robotics and Automation Letters* 8.6 (Jan. 2023), pp. 3740–3747. DOI: 10.1109/LRA.2023.3270034.

[14] Pfanne, M. *In-hand object localization and control : enabling dexterous manipulation with robotic hands*. Springer, 2022. ISBN: 9783031069666.

[15] Ross, S., Gordon, G. J., and Bagnell, J. A. “A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning”. In: (Mar. 2011).

[16] Zhang, J. and Cho, K. “Query-Efficient Imitation Learning for End-to-End Autonomous Driving”. In: (May 2016).